

SUPPLEMENTARY MATERIAL: DETECTING FIRST-PERSON GENDER BIAS IN ENGLISH SPEECH-TO-HINDI TRANSLATION USING WHISPER BASE ASR AND MARIANMT

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0.1. Fairness-Oriented Bias Evaluation

The classified outputs are quantitatively analyzed using fairness-oriented evaluation metrics. The numbers of masculine, feminine, and unknown outputs obtained from the linguistic classifier are used to compute the GBS, GGS, GIS, UR, and ABS and these metrics are defined in 0.2 the next subsection.

These metrics collectively quantify the extent of masculine dominance, gender imbalance, inclusivity, uncertainty, and overall bias present in the generated Hindi translations. The resulting measurements provide a systematic framework for evaluating first-person gender bias in English speech-to-Hindi translation systems.

0.2. Bias Score and Evaluation Metrics

To more accurately evaluate gender bias in first-person English speech-to-Hindi script translation, we extend conventional bias metrics by incorporating an additional category for unknown or neutral outputs. The GBS is defined as the proportion of masculine outputs among all evaluated sentences, while the GGS measures the imbalance between masculine and feminine outputs. To capture inclusive behavior, the GIS measures the proportion of gender-inclusive translations. Additionally, we introduce the UR, which quantifies the proportion of outputs where the model avoids explicit gender assignment. To account for such cases, we further define an ABS, which evaluates bias only among outputs with explicit gender markings. Together, these metrics provide a comprehensive evaluation of both bias and model behavior.

0.2.1. Gender Bias Score (GBS)

We define the Gender Bias Score as:

$$GBS = \frac{\text{Number of masculine outputs}}{\text{Total number of sentences}}$$

where masculine outputs include forms such as रहा (Raha), and जाऊंगा (Jaunga). The total number of sentences includes masculine, feminine, and unknown outputs. This metric captures the extent of male dominance bias.

0.2.2. Gender Gap Score (GGS)

The Gender Gap Score is defined as:

$$GGS = \frac{|N_{male} - N_{female}|}{N}$$

where N_{male} and N_{female} denote the numbers of masculine and feminine outputs, respectively, and N represents the total number of translated sentences.

Unknown outputs are not included in the numerator but are included in the total count. This prevents inflated bias when the model avoids gender assignment.

0.2.3. Gender Inclusivity Score (GIS)

The Gender Inclusivity Score is defined as:

$$GIS = \frac{\text{Number of inclusive outputs}}{\text{Total number of sentences}}$$

where inclusive outputs include forms such as रहा/रही (Raha/Rahi).

0.2.4. Unknown Rate (UR)

The Unknown Rate is defined as:

$$UR = \frac{\text{Number of unknown or neutral outputs}}{\text{Total number of sentences}}$$

Unknown outputs include gender-neutral expressions, ambiguous outputs, or cases with missing verb agreement. High UR indicates that the model avoids making gender decisions. Low UR with high GBS indicates strong male bias. Balanced UR indicates safer but less confident predictions.

0.2.5. Adjusted Bias Score (ABS)

To account for unknown outputs, we define the Adjusted Bias Score as:

$$ABS = \frac{N_{male} - N_{female}}{N - N_{unknown}}$$

where N_{male} , N_{female} , and $N_{unknown}$ denote the numbers of masculine, feminine, and unknown outputs, respectively, and N is the total number of translated sentences. This metric evaluates true bias only among outputs with explicit gender assignments.

0.2.6. Importance of Unknown Handling

Without explicitly accounting for unknown outputs:

- Models may artificially reduce bias by avoiding gendered predictions.
- Apparent bias reduction may be misleading.

The proposed formulation addresses this issue by explicitly modeling unknown outputs. If GBS decreases while UR increases, the model may be avoiding bias rather than correcting it. In contrast, a decrease in GBS accompanied by an increase in GIS indicates effective bias mitigation.

1. EXPERIMENTAL RESULTS

In this section, we reported the dataset description and experimental setup.

1.1. Experimental Setup

Table 1 summarizes the inference configuration employed in the proposed framework. The implementation was carried out in Python using the `whisper`, `transformers`, `edge_tts`, `pandas`, `numpy`, and `pydub` libraries. All experiments were conducted on a system running Windows 11 with 16 GB RAM and an Intel Core i9 processor. As shown in Table 1, speech inputs were generated using Microsoft’s Edge Text-to-Speech (Edge-TTS) system, with the neural voice `hi-IN-SwaraNeural` used for the female speech dataset. The synthesized audio was standardized through mono-channel conversion, resampling to 16 kHz, and waveform normalization before processing. For speech recognition, the Whisper Base model was employed in transcription mode with `fp16=False` to generate English transcripts from speech signals. These transcripts were subsequently translated into Hindi using the MarianMT English-to-Hindi model (`Helsinki-NLP/opus-mt-en-hi`) based on the OPUS-MT architecture and its default generation settings. Finally, the translated Hindi outputs were

Table 1: Inference Configuration

Parameter	Value
Whisper Model	Base
ASR Mode	Transcription
FP16	False
Translation Model	OPUS-MT EN→HI
Audio Channels	1 (Mono)
Sampling Rate	16 kHz
Audio Normalization	Enabled
Gender Classification	Rule-Based

Table 2: Distribution of Gender Labels in Whisper-Generated Hindi Translations

Speech	Masculine	Feminine	Unknown	Total
Female	8160	351	1237	9748
Male	3704	125	184	4013

analyzed using a rule-based gender classification framework that utilizes predefined Hindi linguistic markers to identify masculine, feminine, and unknown forms for subsequent bias evaluation.

1.2. Dataset-Level Gender Distribution Analysis

Table 2 presents the distribution of gender categories identified in the Hindi translations generated by the Whisper-based English-to-Hindi translation system. For female speech inputs, the model generated 8160 masculine outputs compared to only 351 feminine outputs, while 1237 outputs were classified as unknown. Similarly, for male speech inputs, 3704 outputs were classified as masculine and only 125 as feminine. These observations provide preliminary evidence that the translation system strongly favors masculine grammatical constructions regardless of speaker gender. The large disparity between masculine and feminine outputs motivates a deeper quantitative bias analysis.

1.3. Comparative Evaluation Across Speaker Gender

Whisper demonstrates considerable masculine dominance in both speaker situations, according to the comparative study in Table 3. The most alarming finding is the persistence of masculine translations for female speech inputs, despite the fact that masculine outputs are more common for male speech inputs. More than 83% of translations are masculine even when the speaker is female, suggesting that the translation model usually lacks speaker-specific gender cues.

Gender ambiguity may not be handled by the model, as evidenced by the lack of gender-inclusive outputs in

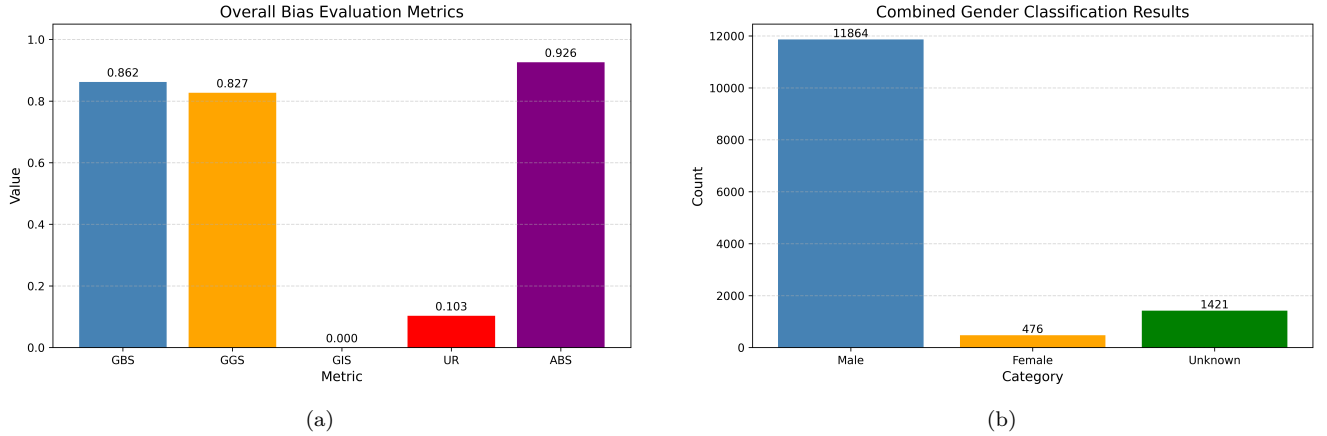


Fig. 1: Overall bias analysis results on the complete dataset. (a) Overall bias evaluation metrics. (b) Combined gender classification results showing a preference toward masculine grammatical forms in Hindi translations.

Table 3: Comparative Bias Evaluation Between Female and Male Speech Inputs

Speech	GBS	GGS	GIS	UR	ABS
Female	0.837	0.801	0.000	0.127	0.918
Male	0.923	0.892	0.000	0.046	0.935

both datasets. Furthermore, even after eliminating unknown outputs, masculine dominance is still significant, as seen by the continuously high Adjusted Bias Scores above 0.91. When taken as a whole, these findings offer compelling empirical proof that systematic first-person male-dominance bias is present in Whisper-based English-to-Hindi translation.

1.4. Evaluation of the Combined Gender Classification Results

The findings of the combined gender classification show a significant imbalance in the translated Hindi outputs, as shown in Fig 1b. Only 476 (3.46%) of the 13,761 translated sentences were categorized as feminine, while 11,864 (86.21%) were categorized as masculine. Due to the lack of clear gender indicators, an additional 1,421 translations (10.33%) were classified as unknown. When translating first-person English utterances into Hindi, the translation pipeline substantially favors masculine grammatical forms, as evidenced by the significant difference between masculine and feminine outputs. This pattern continues even when both male and female speech inputs are included, indicating that the observed bias has its roots in the underlying translation process rather than just speaker traits.

The overall bias metrics further confirm the presence of systematic masculine dominance in the translated out-

puts. As shown in Fig 1a, the GBS of 0.862 indicates that a large proportion of translations were assigned masculine forms. Similarly, the GGS of 0.827 demonstrates a substantial imbalance between masculine and feminine outputs. The GIS of 0.000 indicates the absence of meaningful gender balance in the translated sentences. Although the UR of 0.103 suggests that a portion of translations remained gender-ambiguous, the ABS of 0.926 reveals that the masculine preference remains extremely strong even after excluding unknown cases. Collectively, these results provide strong evidence of systematic masculine bias in the Whisper Base ASR and MarianMT English-to-Hindi translation pipeline, indicating a consistent tendency toward masculine grammatical realization when translating gender-neutral first-person English utterances into Hindi.

2. DISCUSSION

The experimental results reveal a clear presence of first-person gender bias in Whisper based English-to-Hindi translation. Since English expressions such as "I am going," "I am studying," and "I can do this" do not explicitly encode gender, an unbiased translation system should not consistently favor either masculine or feminine Hindi forms. However, the obtained results demonstrate a strong preference for masculine grammatical constructions across both female and male speech inputs. For female speech inputs, the GBS of 0.837 indicates that a large majority of translated sentences were classified as masculine. A significant disparity between masculine and feminine outputs is further supported by the corresponding GGS of 0.801. Even though the model used female speech, it often produced translations that were masculine, such as मैं जा रहा हूँ (Main ja rah hoon)

and मैं पढ़ रहा हूँ (Main padh raha hoon), indicating that speaker-related gender information was not used extensively during translation.

Male voice inputs show a similar tendency, with GBS and GGS values rising to 0.923 and 0.892, respectively. The exceptionally high values show a substantial preference for masculine grammatical forms, even though male speakers are predicted to produce masculine outputs more frequently. After removing unknown outputs, the consistently high ABS—0.918 for female speech and 0.935 for male speech further supports the predominance of masculine translations. The total lack of gender-inclusive outputs is another noteworthy observation. For both datasets, the GIS stays at zero, meaning that the model never produced translations that might maintain gender ambiguity. Instead, it consistently selected a specific gender realization even when the source English utterance provided no explicit gender information. Furthermore, the higher UR for female speech (0.127) compared to male speech (0.046) suggests greater uncertainty in some female-voice translations, although the overall bias pattern remains unchanged.

3. LIMITATIONS

The scope of this study is limited to the detection of gender bias in first-person English-to-Hindi translation. The experiments focus on first-person English utterances, and therefore the findings may not directly generalize to other sentence structures. In addition, the evaluation is conducted using Whisper and the MarianMT English-to-Hindi translation model, and bias characteristics may vary across different translation systems. The proposed framework employs rule-based linguistic markers for gender identification, which are designed for efficient large-scale analysis of translated outputs. Furthermore, the study concentrates on masculine, feminine, and unknown grammatical forms commonly observed in Hindi translations. Future work may extend the framework to additional translation models, broader linguistic contexts, and more comprehensive gender representation analyses.